

Relations & Functions

Relation

In our mathematical language, a relation is a set of ordered pairs, a subset of a Cartesian product

Relation

- If X and Y are sets, then $X \leftrightarrow Y$ denotes the set of all relations between X and Y . The relation symbol may be defined by generic abbreviation

$$X \leftrightarrow Y == \mathbb{P}(X \times Y)$$

Relation

Any element of $X \leftrightarrow Y$ is a set of ordered pairs in which the first element is drawn from X , and the second from Y : that is, a subset of the Cartesian product set $X \times Y$.

Relation Example

Example 7.2 The relation *drives* is used to record which makes of car are driven by the members of a small group of people. If the group of people is defined by

$$\text{Drivers} == \{helen, indra, jim, kate\}$$

and the choice of cars is defined by

$$\text{Cars} == \{alfa, beetle, cortina, delorean\}$$

then *drives* is an element of $\text{Drivers} \leftrightarrow \text{Cars}$, and the statement 'Kate drives a cortina' could be formalised as $kate \mapsto cortina \in drives$. $\square$

Example

Example 7.3 The relation *drives* could be defined by

$$\begin{array}{|l} \text{\textit{_drives_}} : \textit{Drivers} \leftrightarrow \textit{Cars} \\ \hline \textit{drives} = \{ \textit{helen} \mapsto \textit{beetle}, \textit{indra} \mapsto \textit{alfa}, \textit{jim} \mapsto \textit{beetle}, \textit{kate} \mapsto \textit{cortina} \} \end{array}$$

That is, Helen and Jim drive Beetles, Indra drives an Alfa, Kate drives a Cortina, and nobody drives a DeLorean. $\square$

Simple relations can be illustrated using diagrams with arrows, or **graphs**. The graph of *drives* is shown in **Figure 7.1**.

Domain and Range

A relation may contain a great deal of information; often, we require only a small part. To enable us to extract the information that we need, a number of basic functions are included in our mathematical language. The simplest examples are the domain and range functions, 'dom' and 'ran'. If R is a relation of type $X \rightarrow Y$, then the domain of R is the set of elements in X related to something in Y :

$$\text{dom } R = \{x : X; y : Y \mid x \mapsto y \in R \bullet x\}$$

The range of R is the set of elements of Y to which some element of X is related:

$$\text{ran } R = \{x : X; y : Y \mid x \mapsto y \in R \bullet y\}$$

$$\{x : X; y : Y \mid x \mapsto y \in R \wedge x \in A \bullet x \mapsto y\}$$

Example

Example 7.4 The set of people that drive is the domain of *drives*:

$$\text{dom } \textit{drives} = \{\textit{helen}, \textit{indra}, \textit{jim}, \textit{kate}\}$$

The set of cars that are driven is the range:

$$\text{ran } \textit{drives} = \{\textit{alfa}, \textit{beetle}, \textit{cortina}\}$$

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Relation Composition

- if the target type of one relation matches the source type of another then they may be combined to form a single relation.

R is an element of $X \leftrightarrow Y$,

S is an element of $Y \leftrightarrow Z$,

$R \circ S$ to denote the relational composition

This is the element of $X \leftrightarrow Z$ such that

$$x \mapsto z \in R \circ S \Leftrightarrow \exists y: Y \bullet x \mapsto y \in R \wedge y \mapsto z \in S$$

Example-Relation Composition

Example 7.16 The relation *uses* of type *Cars* $\leftrightarrow$ *Fuels* tells us which fuel is used by each of the cars in **Example 7.2**:

uses : *Cars* $\leftrightarrow$ *Fuels*

uses = {*alfa* $\mapsto$ *unleaded*, *alfa* $\mapsto$ *leaded*, *beetle* $\mapsto$ *leaded*,
 cortina $\mapsto$ *leaded*, *delorean* $\mapsto$ *electricity*}

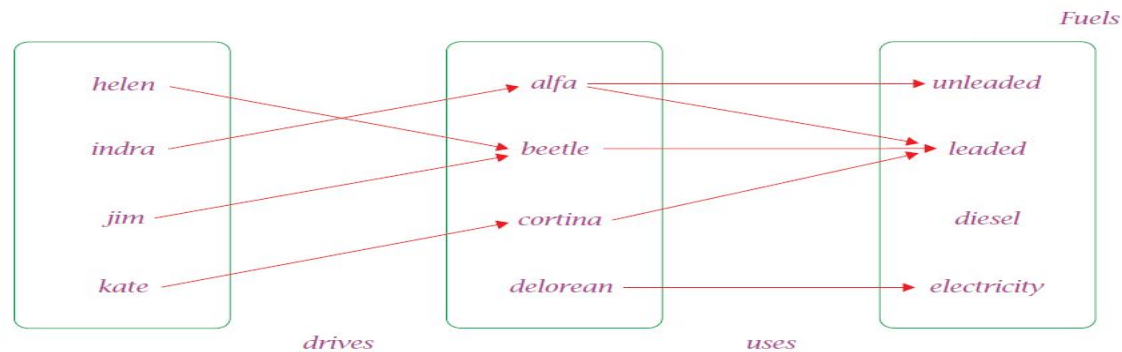


Figure 7.5 Which fuel to buy?

Function

Some relationships can be modelled by a special kind of relation. If each object of one set is related to at most one object of another, then the relation between the two sets is said to be a function: a term that should be familiar from any introduction to mathematics or computation.

Function

A function is a relation which describes that there should be only one output for each input (or) we can say that a special kind of relation (a set of ordered pairs), which follows a rule i.e., every X-value should be associated with only one y-value is called a function

Partial Function

A partial function from X to Y is a relation that maps each element of X to at most one element of Y ; we write $X \rightarrow Y$ to denote the set of all such

$$X \rightarrow Y ==$$

$$\{f : X \leftrightarrow Y \mid \forall x : X; y_1, y_2 : Y \bullet x \mapsto y_1 \in f \wedge x \mapsto y_2 \in f \Rightarrow y_1 = y_2\}$$

Example

Example 8.1 An organisation has a system for keeping track of its employees while they are on the premises. Each employee is issued with an **active badge** which reports their current position to a central database. If the set of all people is *Person*, and the set of all locations is *Location*, then the information provided by the system may be described by a relation *where_is* of type

Person $\leftrightarrow$ *Location*. It is impossible for an employee to be in two places at once, so this relation will be a partial function:

$$\textit{where_is} \in \textit{Person} \rightarrow \textit{Location}$$